

From Web Basics to Digital Makerspaces

ICT Summerschool 2026



**Hochschule
Bonn-Rhein-Sieg**
University of Applied Sciences

**{ FACHBEREICH
INFORMATIK }**

About Me



André Kless, Lecturer

 41 years old

 Father of four children

 Teaching programming, software engineering, and web engineering

 Passionate about making software development accessible to everyone

Who are you?

Course Overview

During this week, you will build your own personal website:

Day	Topic	Outcome
Monday	Web Basics & HTML	Your first website
Tuesday	CSS & Responsive Design	A beautiful website
Wednesday	JavaScript	An interactive website
Thursday	Digital Makerspaces	Embedded web applications
Friday	Data Sovereignty & Web Desktop	Looking beyond today's Web
Online Session 1	Improve your project	Feedback & support
Online Session 2	Final presentation	Showcase your website

Goal: Build and present your own personal website.

Assessment

To successfully complete the course you need to:

- ✓ actively participate
- ✓ build your own personal website
- ✓ present your website during the final online session

Question

Does everyone only need a pass/fail certificate, or does anyone need a graded assessment for their home university or funding programme?

Why the Web?

The Web is the world's largest application platform.



WIKIPEDIA
Celebrating 25 years



They all run in a web browser.

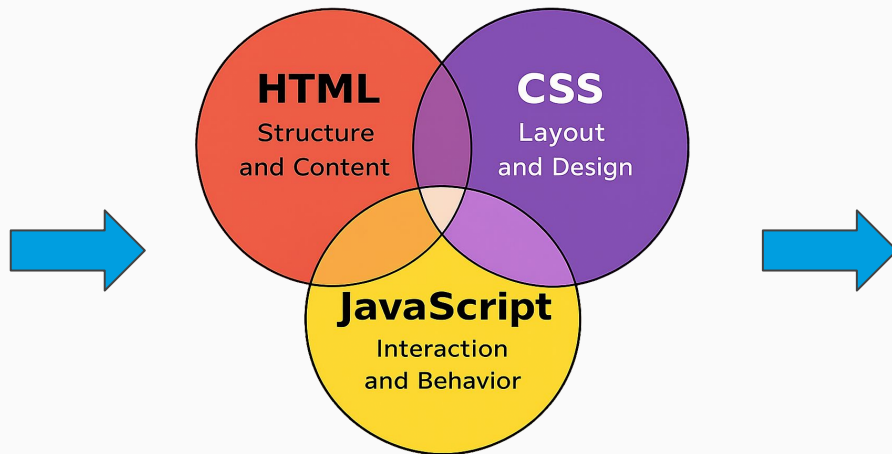
There are many ways to create a website...

Website Builders

- Squarespace
- Webflow
- Wix
- WordPress

AI Assistants

- ChatGPT
- Claude Code
- Gemini
- Lovable
- Bolt



Open Web Standards



HTML

Hypertext Markup Language

Create the structure

- Controls the layout of the content
- Provides structure for the web page design
- The fundamental building block of any web page



CSS

Cascading Style Sheet

Stylize the website

- Applies style to the web page elements
- Targets various screen sizes to make web pages responsive
- Primarily handles the "look and feel" of a web page



Javascript

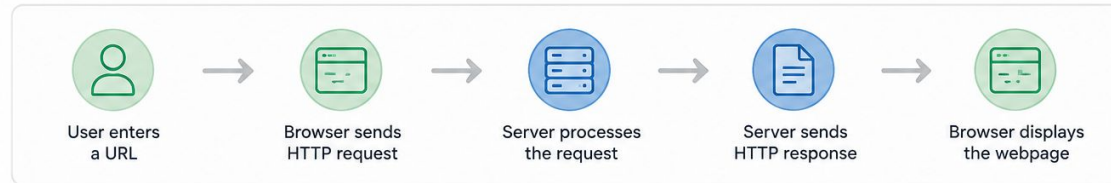
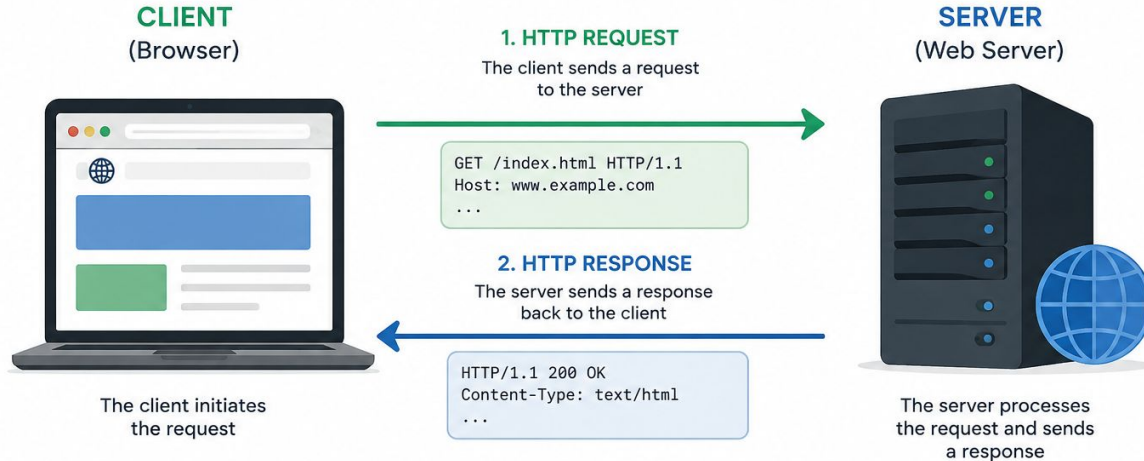
Increase interactivity

- Adds interactivity to a web page
- Handles complex functions and features
- Programmatic code which enhances functionality

How does the Web work?

CLIENT-SERVER ARCHITECTURE

Communication with HTTP



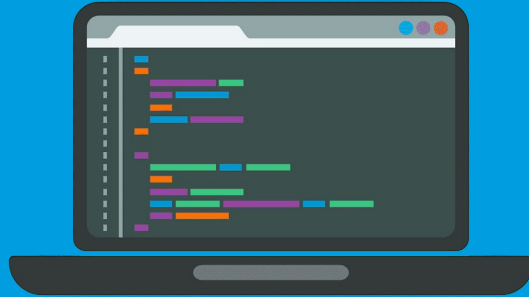
The client (web browser) always starts the communication.

The server only responds.

HTTP is the language browsers and servers use to communicate.

When you open a website, the server responds by sending HTML, CSS, and JavaScript to the browser.

Live Coding



HTML

HTML Recap

What did we learn today?

- ✓ How the Web works
- ✓ How browsers and servers communicate
- ✓ What HTML is
- ✓ How to build your first webpage

Key Takeaways

- HTML describes the structure of a webpage.
- Browsers render HTML into visible content.
- Every website starts with HTML.

Choosing a Code Editor

Editor	Description	Link
Visual Studio Code	Free, cross-platform, beginner-friendly, and extensible.	https://code.visualstudio.com/
WebStorm	Professional IDE for web development (free for students and teachers).	https://www.jetbrains.com/webstorm/
W3Schools Tryit Editor	Browser-based editor – no installation required – perfect for quick experiments.	https://www.w3schools.com/tryit/
Sublime Text	Lightweight, fast, and cross-platform text editor.	https://www.sublimetext.com/
Notepad++	Lightweight text editor for Windows.	https://notepad-plus-plus.org/

Learning Resources

Our primary learning resource:



W3Schools
<https://www.w3schools.com>

Why W3Schools?

- ✓ Beginner-friendly
- ✓ Short explanations
- ✓ Interactive examples
- ✓ Try it yourself in the browser
- ✓ Free to use

Don't be afraid to experiment. That's one of the best ways to learn web development.

Other useful references:



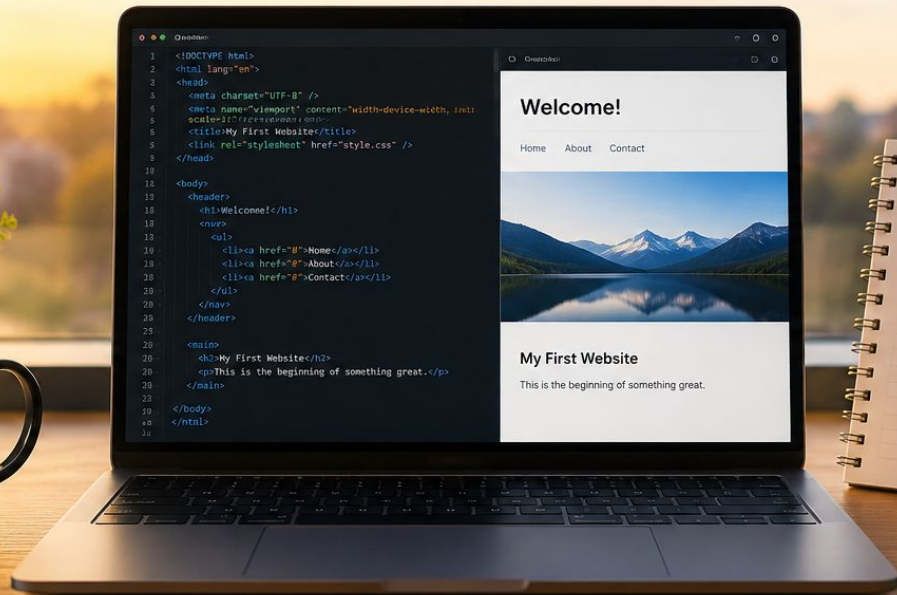
MDN web docs
moz://a

MDN Web Docs
<https://developer.mozilla.org>

Comprehensive reference and documentation for web technologies.

You don't have to remember everything.
You just need to know where to find it.

Let's start!



Today you created...



Let's make it beautiful!